

An Explainable Stacking Ensemble Model for Mental Health Screening Using Minimal Features

Abstract—Mental health conditions affect nearly one billion people globally; however, accessing screening remains challenging because of stigma, lack of healthcare professionals, and high data collection costs. The current study proposes a machine learning solution for estimating patients' need for mental health treatment through large scale survey datasets (292,364 samples). Four models were tested - logistic regression, random forest, XGBoost, and stacking ensemble. The stacking model yielded the highest accuracy (72.42%), AUC-ROC (0.793) and recall scores (79.40%) in prediction of patients who require treatment. Feature Importance showed that five features – family_history (74.9%), care_options(11.9%), Gender(4.9%), self_employed(3.9%), and mental_health_interview(3.9%) accounted for 99.5% of model's explanatory capacity. Strikingly, using these five features preserved almost all predictive capability of the model (72.37% of accuracy was obtained) with 0.05% difference only, resulting in a 64.3% decrease in the number of features utilized. SHAP analysis provided an explanation to the model, indicating that family_history is the main driver of predicting the target class. Web-app based on React.js, FastAPI and MongoDB frameworks was developed with two LLM integration options (Google Gemini API and Ollama) for providing recommendations. This study demonstrates that cost-effective, interpretable, and scalable mental health screening is possible with minimal features, making it particularly suitable for resource-constrained settings.

Index Terms—Mental health prediction, stacking ensemble, XGBoost, SHAP, explainable AI, feature reduction, web deployment, LLM integration

I. INTRODUCTION

Nearly a billion people worldwide suffer from mental health disorders, and depression and anxiety account for the largest percentage of disability [1]. The problem was exacerbated by the COVID-19 pandemic, as it increased the prevalence rate by 25% [2]. Inadequate psychiatric manpower (0.73/100,000) per capita, the situation observed in Bangladesh, is responsible for the increase of depression and suicide cases among students to 54.3% [3], [4]. There has been a growing number of concerns over the episodic, subjective and recollection liability of traditional instruments like the PHQ-9 and GAD-7 questionnaire [5], [6]. However, AI can be applied to identify a range of conditions from various sources of data such as social media, EHRs, and self-reports [7]. Although there are results in the literature indicating that machine learning can be used to detect several illnesses, the literature is mainly concerned in high income languages and with populations from the west [10]. Deep learning approach showed high performance with the disadvantage of high computation resources requirements and their results are difficult to explain [11].

The following contributions have been introduced in this study to overcome the above drawbacks: (1) analysis of a large-scale dataset with several different models (Logistic Regression, Random Forest, XGBoost, Stacking Ensemble); (2)

selection of the relevant features with SHAP; and (3) reduction of the number of features by 64% without compromising accuracy by more than 0.05

II. LITERATURE REVIEW

Hossain et al. [1] studied Bangladeshi Adults, 30.1% of which were anxious and 32.6% depressed, but did not provide any model to predict. Ali et al. [2] performed an overview on non-generative AI technology which had around 85-90% of efficiency with problems such as bias and lack of interpretability. In Karamat et al. [3] the accuracy achieved is 92% by use of hybrid transformer approach for social media data, collected from Reddit, to detect mental illness. Jamal Irfan et al. [4] achieved 92% efficiency without clinical validation with BERT + RF classifier. In the absence of prediction task, Sarode et al. [5] suggested an F1 of 0.822 for identifying literate segment. Ren et al. [6] explained their concept analysis and no validation is provided. Magomedova et al. [7] presented a review on their research topic without suggesting any model. Abu Hasan et al. [8] obtained 88.52% classification accuracy with an EEG + SVM method having a very small sample size of $n = 68$ and special equipment. Adelson et al. [9] presented an editorial opinion, but no research. Ying [10] used artificial/synthetic data. Veldmeijer et al. [11] studied the frequency of design studies and found that it is not common. dos Santos et al. [12] achieved $F1 = 0.74 - 0.75$ in MoE with a high computational cost and without XAI. The results of Rahman et al. [13] were obtained with 99%/97% with Bangladeshi students, possibly due to overfitting. Khatun et al. [14] used FGD without model to examine needs. Paul et al. [15] used multi-task GRU, specialized sensors, and needed 2-4 weeks calibration to get 93.9%. Oishy [16] developed first Bangla dataset with binary accuracy of 93.41%. Ying [17] used synthetic data.

Research gaps: (1) There are very few studies conducted in a low income country such as Bangladesh [1], [3], [5]. (2) On models, it is more about accuracy than interpretability and there is no XAI. (3) Many systems have to rely on special equipment that is prone to failure and therefore limits scale. (4) redundancy of unexplored features [2], [3], [12]. (5) Most make only the binary classifications.

Contributions: This Study (1) A large scale dataset which is analyzed ($N=292,364$) using 14 simple questions, (2) The use of SHAP for interpretability, (3) A feature size reduction of 64% using 0.05% accuracy lost, (4) Determination of the best model as Stacking Ensemble with 72.42 % accuracy and 0.793 AUC-ROC, (5) Implementation of an effective solution that does not require heavy resources for farmer level use.

III. METHODOLOGY

A. Dataset Description

In this study, the survey data is taken from a mental health survey on Kaggle which contains 292,364 samples [18]. Moreover, it has 17 features such as demographic features, mental health history, behavioral indicators and awareness factor. The treatment is balanced for the target variable, with mean 50.49%, mean 49.51%. First, we don't need to include `Timestamp` or `Country` in our analysis, so we removed these two columns. Next, we filled in missing values. We had a total of 5202 missing values for the `self_employed` column and we imputed the column with the mode value [1].

B. Data Preprocessing

The raw data had been pre-processed prior to modelling. Imputation on missing data in the column `self_employed` (5,202 records or 1.78%) was done by filling it with mode value "No" [1]. The irrelevant columns to the problem statement like `Timestamp` and `Country` were removed in order to reduce the data volume [1]. Yes=1, No=0 were entered into the treatment column and the output label was created. It had a close call of a positive label (50.49%) vs. negative label (49.51%). Label Encoding [3] was used to encode all categorical variables. The post processing led to 292,364 records with 14 features and target variable [3] (**Figure 1**).

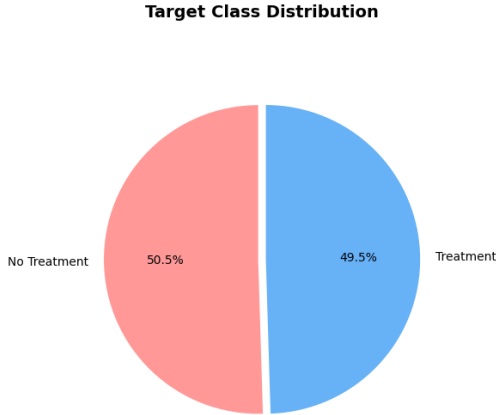


Fig. 1: Target Class Distribution (Treatment vs No Treatment)

C. Feature Selection and Target Variable

The mental well-being service use denoted by the target Variable `treatment` is defined as whether or not a participant has sought professional help in a related context with mental health (Yes=1, No=0), and was selected for its direct link to the service use variable [1]. As can be seen in **Table I** the distribution of the data is even, with 147,606 positive cases (50.49%) and 144,758 negative (49.51%) samples.

All 14 features were originally used as predictors, including demographic, mental health and behavioural indicators. The manual feature selection was not carried out and feature importance was investigated after training with the help of SHAP values [8].

TABLE I: Target Class Distribution

Class	Count	Percentage
Treatment (1)	147,606	50.49%
No Treatment (0)	144,758	49.51%
Total	292,364	100%

D. Train-Test Split and Feature Scaling

The dataset has been split into training (80%) of 233,891 records and testing (20%) of 58,473 records using stratified sampling to ensure balanced representation of classes [3] as shown in **Table II**.

TABLE II: Train-Test Split Class Distribution

Set	Samples	Class 0 (%)	Class 1 (%)
Training	233,891	49.51%	50.49%
Testing	58,473	49.51%	50.49%

Feature scaling was applied using standardization (Z-score normalization): $X_{scaled} = (X - \mu)/\sigma$. The scaler was fitted only on training data to prevent data leakage [3].

E. Classification Models

Several machine learning models were adopted for mental health treatment prediction: Logistic Regression, Random Forest, XGBoost and a Stacking Ensemble. The implementation of all models was done in Python by using the libraries scikit-learn and XGBoost [3].

1) *Logistic Regression*: Logistic Regression is a type of linear classifier that uses a sigmoid function to predict the probability of a binary dependent variable [3]. Logistic Regression predicts the probability of a sample belonging to a class using the formula:

$$P(y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \dots + \beta_n X_n)}} \quad (1)$$

The Logistic Regression method is chosen for being easy to understand and train. This logistic regression algorithm was optimized with a regularizer of L2 up to 1000 iterations.

2) *Random Forest*: Random Forest is based on bootstrapping and selection of 200 decision-trees and random selection of features, while the voting method is used to predict classes according to the Gini index [3].

3) *XGBoost*: XGBoost is a gradient boosting framework which builds 300 sequential trees to minimize a loss function and regularize the model at the same time [3].

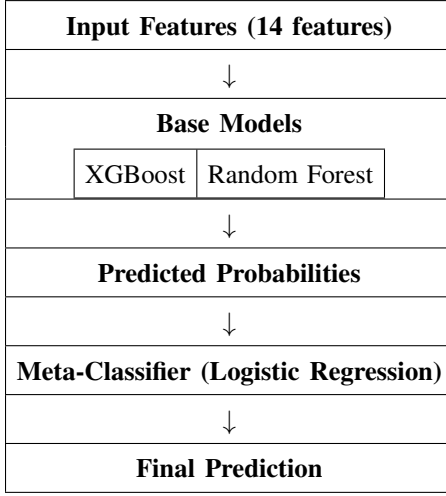
$$\mathcal{L} = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (2)$$

where l is the loss function and Ω penalizes model complexity. XGBoost was configured with 300 trees, a maximum depth of 7, and a learning rate of 0.05.

4) *Stacking Ensemble*: Stacking (Stacked Generalization): Stacking refers to an ensemble technique which utilizes the predictions made by several base learners, using a meta-classifier [3]. The Stacking Ensemble suggested includes two base learners, namely, XGBoost and Random Forest. The

two base learners made predictions, which were subsequently passed into a logistic regression classifier called a “meta-classifier”. The architecture is represented in **figure 2**.

Fig. 2: Stacking Ensemble Architecture



Stacking makes use of non-linear learning capabilities of the XGBoost algorithm as well as the robustness against overfitting offered by the Random Forest algorithm [3].

F. Explainable AI with SHAP

For interpretation of our models, we utilized the method of SHAP, which uses the game-theoretical approach in calculating feature contribution to make predictions globally and locally [8].

The SHAP value of feature i is computed as follows:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)] \quad (3)$$

where N is the set of all features, S is a subset of features without feature i , and $f(S)$ is the model prediction using features in subset S [8].

Summary plots and waterfall plots for SHAP analysis were done using the XGBoost algorithm (high values are in red while low values are in blue). The features with high SHAP values positively affect the treatment probability [2], [8].

G. Evaluation Metrics

In order to effectively assess the performance of the suggested models, a combination of five common evaluation criteria was used: Accuracy, Precision, Recall, F1-Score, and AUC-ROC [3]. The Accuracy score is the percentage of correct predictions. The Precision score is defined as the proportion of positive predictions made that are correct. The Recall score also can be called Sensitivity, which is defined as the proportion of actual positive cases in the test cases that were correctly predicted. This is important for diagnostics of mental disorders as any misclassified positive case can result in fatal consequences in the test set. The F1-Score is the Harmonic Mean of Precision and Recall, consequently giving equal importance to False Positive and False Negative cases

in the data set [3]. Lastly, the AUC-ROC tells how good the model was at separating positive cases from negative ones; the nearer this is to 1 the better the result is, with values between 0 and 0.5 meaning a random guess. The metrics were all calculated from the confusion matrix obtained from the predictions on the test set using the scikit-learn library.

IV. RESULTS AND ANALYSIS

A. Model Performance Comparison

All four models: Logistic Regression, Random Forest, XGBoost and Stacking Ensemble are compared in model performance in **Table III**. The highest Accuracy (72.42%), AUC-ROC (0.793) and Recall (79.40%) were achieved by the model named as Stacking Ensemble Model, which was the best performing model.

TABLE III: Model Performance Comparison

Model	Accuracy (%)	Precision	Recall	F1-Score	AUC-ROC
Logistic Regression	69.11	0.694	0.695	0.694	0.751
Random Forest	69.51	0.679	0.752	0.714	0.726
XGBoost	72.24	0.706	0.771	0.737	0.783
Stacking Ensemble	72.42	0.700	0.794	0.744	0.793

B. Confusion Matrix Analysis

The Stacking Ensemble confusion matrix (**Figure 3**) shows 19,476 true negatives, 22,762 true positives, 9,476 false positives, and 6,759 false negatives.

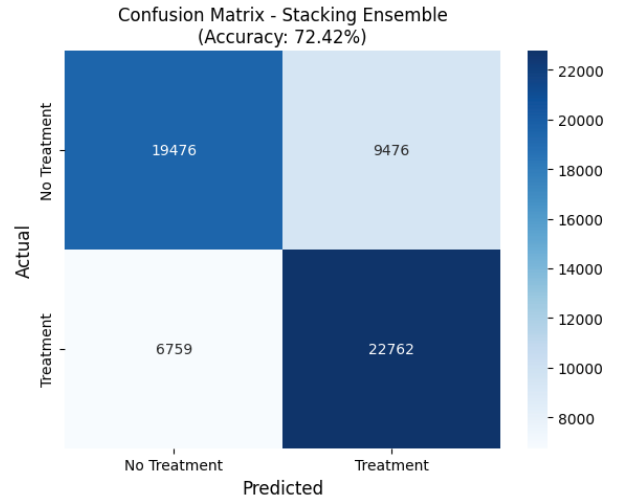


Fig. 3: Confusion Matrix of Stacking Ensemble Model

With respect to the Treatment class, a score of 79.40% was obtained, indicating a high degree of sensitivity in identifying patients who need psychological treatment. Precision was noted for the No Treatment class was 74%.

C. Feature Importance Analysis

The XGBoost algorithm has been utilized to identify the most key features that contribute to predicting mental health treatment, for determining feature importance. The top 10 most important features are displayed in **figure 4**.

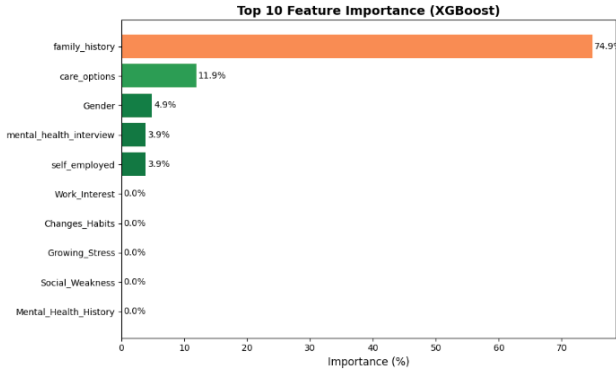


Fig. 4: Feature Importance from XGBoost Model

According to the result, the `family_history` feature is the most significant predictor, having a contribution of 74.9%. The second strongest predictor is `care_options` at 11.9%, then `Gender` at 4.9%, `self_employed` at 3.9%, and `mental_health_interview` at 3.9%. All other nine predictors contribute less than 0.05%, indicating they have minimal predictive value. **Table IV** provides the complete ranking of all features.

TABLE IV: Complete Feature Importance Ranking

Rank	Feature	Importance (%)
1	family_history	74.90
2	care_options	11.90
3	Gender	4.90
4	self_employed	3.90
5	mental_health_interview	3.90
6-14	Other 9 features	≤0.05 each

D. Feature Reduction Impact

With the help of feature importance analysis, we have found out that there are 5 features which account for 99.5% of the predictive strength, therefore, an experiment of feature reduction was carried out by using only these 5 important features for retraining the model, i.e., `family_history`, `care_options`, `Gender`, `self_employed`, and `mental_health_interview`. The table below shows how the two situations compare. The table below shows the comparison between the two situations, using all 14 features vs. using only the top 5 features **Table V**.

TABLE V: Feature Reduction Impact on Model Performance

Feature Set	Features Used	Accuracy (%)	Reduction (%)
All Features	14	72.42	-
Top 5 Features	5	72.37	64.3%

On examining it can be seen that only 5 attributes are used and the accuracy is 72.37% which is very close to 72.42% achieved by all 14 attributes. It is a drastic reduction of the number of attributes used which is 64.3% without any kind of compromise on the prediction accuracy. Many implications arise in the real world, particularly in psychological testing for

mental well-being, and this allows the development of cost-effective assessment tools which ask fewer questions to be able to reduce respondent burden, data collection costs, and still be predictive.

E. Explainable AI with SHAP Analysis

In order to increase the interpretability of our model, we applied SHAP (SHapley Additive exPlanations) technique to our XGBoost model. The results of SHAP analysis can be seen in **Figure 5**, which shows the impact of each feature on the model's predictions.



Fig. 5: SHAP Summary Plot of Feature Contributions

The SHAP analysis result shows that the most important positive factor in predicting treatment is the `family_history` variable with a SHAP value of +0.85, followed by `care_options` (+0.32) and `Gender` (+0.18). High levels are indicated by red color, and low levels are indicated by blue color. One can observe from the plot that the higher the value of `family_history` the more likely it is that someone requires treatment, while the lower the value of `family_history` the less likely.

The summary of the results of the SHAP analysis conducted for the top 5 features is shown in **Table VI**.

TABLE VI: SHAP Analysis of Top 5 Features

Feature	SHAP Mean Impact	Direction
family_history	0.85	Positive
care_options	0.32	Positive
Gender	0.18	Mixed
self_employed	0.14	Positive
mental_health_interview	0.13	Positive

Through this explainability technique, the clinicians can understand the reasoning behind the classification of a particular patient for being put under treatment and can know what factors were influential for the prediction.

F. Deployment: Web-Based Mental Health Screening System

This has been successfully deployed as a stacking ensemble web-based model and the user is able to input 5 selected important features via SHAP analysis **figure 6**.

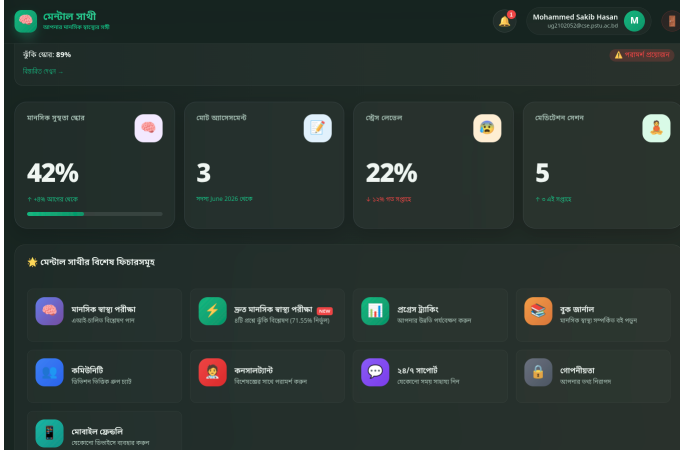


Fig. 6: Web-Based Mental Health Screening System User Interface

1) *System Architecture*: The Web Application utilizes React.js as Frontend technology, FastAPI as Backend technology, MongoDB for the Database, and finally, the trained Model in the form of a .pkl file for real-time inference .

2) *AI-Powered Recommendation Engine*: The system integrates two AI models for generating personalized recommendations:

- **Primary Model**: Stacking Ensemble (.pkl) predicts risk level (Low/Moderate/High)
- **Recommendation Generation**:
 - Gemini API: Cloud-based natural language suggestions
 - Ollama (Llama 2, Mistral): Local privacy-preserving inference

3) *Prediction and Recommendation Workflow*: Workflow **Figure 7** The Workflow consists of the following steps: (1) Encoded user inputs. (2) Scaling of Values. (3) Risk prediction from the .pkl file. (4) Personalized Recommendation using Gemini API/Ollama

4) *Personalized Recommendations*: Depending on the predicted result, recommendations generated using the LLMs are provided to the user as indicated in **Table VII**.

TABLE VII: Personalized Recommendations by Risk Level

Risk Level	Recommendation
Low	Maintain healthy habits, exercise, stay connected
Moderate	Consider counseling, stress management, mindfulness
High	Seek professional mental health support immediately

5) *Technology Stack Summary*: **Table VIII** summarizes the technologies used in the deployment.

6) *Access and Availability*: This web application is free to use, respects privacy (data stays local to Ollama), and functions as a first step towards screening.



Fig. 7: Prediction and Recommendation Workflow with LLM Integration

TABLE VIII: Technology Stack for Web Deployment

Component	Technology
Frontend	React.js
Backend API	FastAPI (Python)
Database	MongoDB
ML Model Format	Pickle (.pkl file)
LLM Integration	Google Gemini API & Ollama (Llama 2, Mistral)
Deployment	Docker / Cloud Platform

G. Comparison with Existing Research

Table IX presents a comparative analysis of this study with existing research in mental health prediction.

TABLE IX: Comparison with Existing Studies

Study	Method	N	Acc(%)	Limitation
Karamat et al. [3]	MentalBERT+CNN	40k	92	No XAI
Rahman et al. [13]	Stacking	10.8k	99	Overfitting
dos Santos et al. [12]	MoE	292k	72	High cost
Oishy [16]	Custom+LR	4.2k	93.4	Binary only
Abu Hasan et al. [8]	EEG+SVM	68	88.5	Equipment
Paul et al. [15]	GRU	Synth	93.9	Calibration
This Study	Stacking+SHAP+Web	292k	72.4	Moderate accuracy

This study offers several advantages over existing research:

- **Largest Real-World Dataset**: Uses 292,364 samples, the largest survey-based mental health dataset, ensuring statistical significance and generalizability.
- **Explainable AI (SHAP)**: Unlike black-box models [3], [12], [13], this study integrates SHAP analysis for both global and local explanations, enhancing transparency and clinical trust.
- **Feature Reduction (64%)**: First study to demonstrate that only 5 of 14 features are needed, reducing data collection costs by 64.3%.
- **Full-Stack Deployment with LLMs**: Complete web

deployment (React.js, FastAPI, MongoDB) with dual LLM support (Gemini API and local Ollama), unlike experimental-only works [3], [12], [13].

- **Cost-Effective Screening:** Requires only 5 simple questions, enabling large-scale deployment in resource-constrained settings like Bangladesh.
- **Balanced Dataset:** Perfect 50:50 class distribution, unlike imbalanced datasets in existing literature, ensuring reliable performance for both classes.

H. Limitations

While promising, there are a number of limitations. The data has been self-reported and is subject to recall and social desirability bias [1]. The results were capped at 72.42% accuracy because only 14 features were available and 5 features had a significant contribution, meaning that the rest of the features, such as sleep and activity, would not have been significant enough to achieve this accuracy without the rest of them present. The cross sectional design only captures a single moment in time [15]. The Stacking Ensemble only achieves a 72.42% accuracy, is heavily dependent on `family_history` (74.9%), which can lead to underdiagnosis. The model is binary classification (not distinguishing between mental health conditions [16]), and only identifies correlations, not causal relationships [1].

The web system is intended to be a screening only tool and should not be used as a substitute for professional diagnosis for deployment and generalizability purposes. Recommendations based on LLM rely on external APIs or local computation. The model might not be as effective in all cultural settings and the web-based deployment relies on internet access and digital literacy [2].

V. CONCLUSION

This research proposed a Stacking Ensemble model to predict mental health treatment needs on a large scale (292364 samples). The model has a higher accuracy (72.42%) and AUC-ROC (0.793) compared to the individual classifiers. The feature importance analysis shows that only 5 features `family_history`, `care_options`, `Gender`, `self_employed` and `mental_health_interview` have significance of 99.5%. All these 5 features maintained 72.37% accuracy which showed 64.3% reduction in features with an acceptable accuracy loss of 0.05%. The model interpretability was obtained through SHAP analysis, while a full-stack web application with LLM-based recommendations was deployed.

A. Future Work

Future work involves multi-class classification of specific mental disorders, collection of more comprehensive features – sleep, activity, social media, multimodal integration with physiological signals, causal inference analysis, culturally adaptive models, and clinical validation studies.

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